Correlation Analysis of the Abundance of Three Dominant Benthic Species Along the Schleswig-Holstein Coast of the Baltic Sea

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Abstract

Benthic communities play an important role in marine ecosystems, and the interspecific interactions among species have significant impacts on community structure and ecosystem functioning. Based on multi-year monitoring data from the Baltic Sea coast of Schleswig-Holstein, Germany, this study systematically analyzes the abundance correlations among three dominant benthic organisms: the blue mussel (*Mytilus edulis*), a filamentous red alga (*Polysiphonia* sp.), and the barnacle (*Amphibalanus improvisus*). Using R language, 159 samples were analyzed through descriptive statistics, Pearson linear correlation, Spearman and Kendall rank correlations, partial correlation analysis, and linear regression modeling, with various visualization methods employed to display interspecific relationships. Results showed that the three species had mean percent covers of 38.17%, 12.33%, and 9.23%, respectively, with coefficients of variation all exceeding 100%, indicating highly dispersed distributions. Pearson correlation analysis revealed a moderate negative correlation between mussels and *Polysiphonia* ($r = -0.30$, $p < 0.001$), while the Spearman rank correlation coefficient was higher ($\rho = -0.44$, $p < 10^{-8}$), suggesting a nonlinear competitive relationship between the two. Correlations between mussels and barnacles, and between *Polysiphonia* and barnacles were weak ($|r| < 0.20$). Partial correlation analysis showed that after controlling for the third species, the negative correlation between mussels and *Polysiphonia* remained significant (partial correlation coefficient -0.28, $p = 0.00036$), confirming a direct competitive effect between them. This study provides quantitative evidence for ecological management and monitoring of benthic communities in the Baltic Sea and offers a reproducible analytical framework for future research.

Keywords: benthic organisms; correlation analysis; Baltic Sea; *Mytilus edulis*; *Polysiphonia* sp.; *Amphibalanus improvisus*; R language; ecology

1 Introduction

1.1 Research Background

The Baltic Sea is one of the world's largest semi-enclosed brackish water bodies, and its unique environmental conditions have nurtured rich benthic communities. As an important component of marine ecosystems, benthic organisms play irreplaceable roles in nutrient cycling, energy flow, water purification, and habitat creation [5]. Along the Baltic Sea coast of Schleswig-Holstein, *Mytilus edulis*, *Polysiphonia* sp., and *Amphibalanus improvisus* are dominant species that form typical sessile organism communities on hard substrates.

These three organisms exhibit some overlap in their ecological niches, as they all require attachment to hard substrates for growth and depend on nutrients and plankton in seawater. Mussels are bivalve mollusks that attach to rocks and other hard substrates via byssal threads and feed by filtering phytoplankton and organic debris; *Polysiphonia* is a filamentous red alga capable of attaching to various hard substrate surfaces for photosynthesis; barnacles are crustaceans whose larvae float in seawater for a period before selecting suitable substrates to attach and develop into adults.

In recent years, global climate change, ocean acidification, eutrophication, and human activities such as shipping, fisheries, and coastal development have significantly impacted the Baltic Sea ecosystem. Temperature increases have altered species growth rates and reproductive cycles, salinity fluctuations have affected species' physiological adaptability, and changes in nutrient input have led to alterations in primary productivity. These environmental changes are modifying the structure of benthic communities and the interspecific interactions, thereby affecting the stability and functioning of the entire ecosystem.

1.2 Theoretical Foundation of Interspecific Interactions

Species in biological communities do not exist in isolation but are interconnected through various mechanisms such as competition, predation, and symbiosis. Competition is one of the most fundamental interspecific relationships in communities; it occurs when the demand of two or more species for the same resource (such as space, food, or light) exceeds the resource supply. According to niche theory, the higher the niche overlap, the more intense the competition. In sessile organism communities, substrate space is one of the most important limiting resources. When mussels attach in large numbers and form dense mussel beds, they occupy substantial space and limit the colonization of other species; similarly, abundant growth of *Polysiphonia* may cover substrate surfaces, forming algal mats that affect the attachment of other organisms.

Interspecific interactions can be quantified through correlation analysis. Positive correlation may indicate that two species have similar ecological requirements or mutualistic relationships, while negative correlation suggests the presence of competition or other negative interactions. However, correlation analysis can only reveal the strength and direction of associations and cannot directly prove causal relationships. Partial correlation analysis can assess the net correlation between two variables while controlling for the influence of other variables, which helps understand direct interspecific interactions.

1.3 Research Status and Gaps

Domestic and international scholars have conducted extensive research on Baltic Sea benthic organisms. Many studies focus on the ecological characteristics of single species, and some have explored the overall structure and diversity of benthic communities, but these studies are often based on short-term observations or localized areas, lacking support from long-term time series and large spatial scale data.

Quantitative research on interspecific interactions is relatively scarce. Although some experimental studies have explored competition and mutualistic relationships among species under controlled conditions, analysis of field observation data remains insufficient. In particular, there is a lack of systematic quantitative assessment of the correlations among these three dominant species: mussels, *Polysiphonia*, and barnacles. Additionally, existing research mostly employs traditional statistical methods and rarely utilizes modern computational tools and visualization techniques to comprehensively display and interpret complex ecological data.

1.4 Research Objectives and Significance

This study aims to systematically analyze the abundance correlations among three dominant benthic organisms along the Schleswig-Holstein coast of the Baltic Sea using long-term monitoring data. Specific objectives include:

1. Describe the abundance distribution characteristics of the three benthic organisms, including means, variability, and spatial distribution patterns;

2. Apply multiple correlation analysis methods (Pearson, Spearman, Kendall) to comprehensively evaluate linear and nonlinear relationships among species;

3. Through partial correlation analysis, eliminate the influence of the third species to reveal direct interactions between species pairs;

4. Use linear regression models to quantitatively describe relationships among species abundances and perform residual diagnostics;
5. Employ various visualization methods (histograms, boxplots, scatterplots, heatmaps, etc.) to intuitively display analysis results;
6. Based on analysis results, explore possible ecological mechanisms and provide scientific evidence for monitoring and management of benthic communities.

The significance of this study lies in: first, filling the gap in quantitative research on interspecific interactions among Baltic Sea benthic organisms; second, providing a complete, reproducible data analysis workflow that offers methodological reference for similar studies; third, providing scientific evidence for marine ecosystem management and conservation, helping predict the impacts of environmental changes on benthic communities.

2 Materials and Methods

2.1 Study Area

The study area is located along the Baltic Sea coast of Schleswig-Holstein in northern Germany. This region has a temperate maritime climate with an average annual temperature of approximately 8–10°C and annual precipitation of 700–800 mm. The Baltic Sea here has relatively low salinity (approximately 7–12 psu), ranging between freshwater and seawater, forming a unique brackish water environment. The coastal substrate consists mainly of rocks, gravel, and shells, providing good habitat for sessile benthic organisms.

2.2 Data Source

The data used in this study come from the `Benthic_species` monitoring dataset, which records percent cover of benthic organisms at multiple sites along the Baltic Sea coast over multiple years of monitoring. The dataset contains 159 samples, each recording the coverage of four benthic organisms. Percent cover refers to the percentage of substrate surface area occupied by a species and is a commonly used metric for measuring the abundance of sessile organisms.

According to the research objectives, this paper focuses on analyzing three dominant species: mussels *Mytilus edulis*, filamentous red alga *Polysiphonia* sp., and barnacles *Amphibalanus improvisus*. These three organisms occupy important positions in the benthic community of this region, and their interactions have significant impacts on community structure.

2.3 Data Preprocessing

All data processing and statistical analyses were completed in the R language environment [1]. Data preprocessing steps included:

1. **Data Reading:** Use the `read.table()` function to read tab-delimited text files, specifying headers and delimiters;
2. **Variable Selection:** Extract the `Mytilus`, `Polysiphonia`, and `Balanus` columns from the raw data;
3. **Variable Renaming:** Rename columns to `Mytilus_edulis`, `Polysiphonia_sp`, and `Amphibalanus_improvisus` to clarify species scientific names;
4. **Data Quality Check:** Check for missing values, outliers, and data ranges. Upon inspection, there were no missing values in the data, and all coverage values were between 0–120% (some sampling points may exceed 100% due to multi-layer coverage);
5. **Data Structure Confirmation:** Confirm that data are numeric and sample size is 159.

All analyses were based on original coverage values without data transformation or standardization to maintain the ecological significance and interpretability of percent cover.

2.4 Descriptive Statistics

Descriptive statistical analyses were performed on the coverage data of the three benthic organisms, calculating the following indicators:

- **Central Tendency:** Mean and median
- **Dispersion:** Standard deviation, minimum, and maximum values
- **Coefficient of Variation:** CV = (SD / mean) × 100%, used to compare the relative degree of variation in coverage among different species

2.5 Correlation Analysis

Three correlation analysis methods were employed to comprehensively assess associations among species:

2.5.1 Pearson Correlation Coefficient

The Pearson correlation coefficient (r) measures the degree of linear correlation between two variables, with values ranging from $[-1, 1]$. The calculation formula is:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{1}$$

where x_i and y_i are the i-th observed values of the two variables, $\bar{x}$ and $\bar{y}$ are means, and n is sample size. Pearson correlation is suitable for normally distributed continuous variables and is sensitive to outliers.

2.5.2 Spearman Rank Correlation Coefficient

The Spearman rank correlation coefficient (ρ) is a nonparametric correlation method that calculates Pearson correlation after performing rank transformation on the data. This method has no requirements for data distribution, is insensitive to outliers, and is suitable for nonlinear monotonic relationships. The calculation formula is:

$$\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2 - 1)} \tag{2}$$

where d_i is the difference in ranks of the i-th observation between the two variables.

2.5.3 Kendall Rank Correlation Coefficient

The Kendall rank correlation coefficient (τ) is also a nonparametric method based on the proportion of concordant and discordant pairs. The calculation formula is:

$$\tau = \frac{n_c - n_d}{\frac{1}{2}n(n-1)} \tag{3}$$

where n_c is the number of concordant pairs and n_d is the number of discordant pairs. Kendall correlation is more robust for data with smaller sample sizes.

Two-tailed significance tests were performed for each species pair, with significance level set at $\alpha = 0.05$. The null hypothesis tested is: there is no correlation between the two variables ($H_0 : \rho = 0$).

2.6 Partial Correlation Analysis

Partial correlation analysis is used to assess the net correlation between two variables while controlling for the influence of other variables. For three variables X_1, X_2, and X_3, the partial correlation coefficient of X_1 and X_2 after controlling for X_3 is defined as:

$$r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{(1 - r_{13}^2)(1 - r_{23}^2)}} \tag{4}$$

where r_{12}, r_{13}, and r_{23} are the simple correlation coefficients between variable pairs, respectively.

This study uses the `ppcor` package [2] in R to calculate partial correlation matrices and corresponding p-values to identify direct interactions between species pairs.

2.7 Linear Regression Analysis

To further quantify relationships among species, simple linear regression models were established for each species pair:

$$Y = \beta_0 + \beta_1 X + \epsilon \tag{5}$$

where Y is the dependent variable (response species), X is the independent variable (predictor species), β_0 is the intercept, β_1 is the slope, and ϵ is the random error term.

The following diagnostics were performed for each model:

- **Goodness of Fit:** Calculate the coefficient of determination R^2 and adjusted R^2;
- **Parameter Significance:** Test whether the slope β_1 is significantly different from zero;
- **Residual Diagnostics:** Plot residual plots, QQ plots, and residuals vs. fitted values plots to test normality, linearity assumptions, and homoscedasticity.

2.8 Data Visualization

Various graphics were used to display data distributions and interspecific relationships:

- **Histograms:** Display frequency distributions of coverage for each species;
- **Boxplots:** Compare distribution characteristics and outliers of coverage for the three species;
- **Kernel Density Curves:** Smoothly display probability distributions of coverage;
- **Scatterplot Matrices:** Display pairwise scatter relationships among species;
- **Correlation Heatmaps:** Use colors to represent the magnitude and direction of correlation coefficients;

- **Regression Scatterplots:** Display linear regression fit lines and confidence intervals.

All graphics were created using R packages including `ggplot2`[3], `corrplot`[4], and `GGally`, ensuring aesthetic quality and professionalism of the figures.

2.9 Software and Tools

The main software and R packages used in this study include:

- R language (version 4.x) [1]: primary analysis platform;
- `ggplot2`[3]: data visualization;
- `corrplot`[4]: correlation matrix visualization;
- `GGally`: scatterplot matrices;
- `ppcor`[2]: partial correlation analysis;
- `reshape2`: data reshaping;
- `RColorBrewer`: color schemes.

All analysis code and generated charts are saved in the respective directories (`results/` and `figures/`), ensuring reproducibility and transparency of the research.

3 Results

3.1 Descriptive Statistics Results

Descriptive statistics results for the coverage of the three benthic organisms are shown in Table 1. Mussels had the highest mean coverage at 38.17%, with a standard deviation of 41.85% and coefficient of variation of 109.65%. The median was 15%, much lower than the mean, indicating a right-skewed distribution with some sites having particularly high coverage. Mussel coverage ranged from 0–120%, with values exceeding 100% possibly due to multi-layer attachment.

Polysiphonia had a mean coverage of 12.33%, standard deviation of 22.24%, and coefficient of variation as high as 180.44%, showing higher spatial heterogeneity. The median was 0, indicating that more than half of the sampling sites had absent or very low *Polysiphonia* coverage, but where present, it could form relatively high coverage, with a maximum value of 100%.

Barnacles had the lowest mean coverage at 9.23%, standard deviation of 17.87%, and the highest coefficient of variation at 193.59%. The median was 1%, with coverage ranging from 0–90%. The coefficients of variation for all three species exceeded 100%, indicating highly dispersed coverage in space, reflecting the patchy distribution characteristics of benthic organisms.

Table 1: Descriptive statistics of percent cover for three benthic organisms ($n = 159$)

Species	Mean(%)	SD(%)	Median(%)	Min(%)	Max(%)	CV(%)
Mytilus edulis	38.17	41.85	15	0	120	109.65
Polysiphonia sp.	12.33	22.24	0	0	100	180.44
Amphibalanus improvisus	9.23	17.87	1	0	90	193.59

3.2 Correlation Analysis Results

3.2.1 Pearson Linear Correlation

The Pearson correlation coefficient matrix for the three benthic organisms is shown in Table 2. Mussels and *Polysiphonia* exhibited a moderately strong negative correlation ($r = -0.30$), indicating a possible competitive relationship between them. Mussels and barnacles showed a weak negative correlation ($r = -0.158$), while *Polysiphonia* and barnacles showed a weak positive correlation ($r = 0.173$).

Table 2: Pearson correlation coefficient matrix

	Mussel	*Polysiphonia*	Barnacle
Mussel	1.0000	−0.3000	−0.1584
Polysiphonia	−0.3000	1.0000	0.1730
Barnacle	−0.1584	0.1730	1.0000

3.2.2 Spearman and Kendall Rank Correlations

Table 3 shows the Spearman and Kendall rank correlation coefficients. Compared to Pearson correlation, the Spearman correlation coefficient showed a stronger negative correlation. The Spearman rank correlation coefficient between mussels and *Polysiphonia* reached −0.44, and the Kendall coefficient was −0.33, both significantly higher than the absolute value of the Pearson correlation coefficient. This indicates that the negative correlation between them is more pronounced at the rank level, possibly suggesting a nonlinear competitive effect.

The rank correlation coefficients between mussels and barnacles were close to zero ($\rho = -0.017$, $\tau = -0.010$), much lower than the Pearson correlation coefficient, indicating a weak linear relationship. The rank correlation coefficients between *Polysiphonia* and barnacles ($\rho = 0.139$, $\tau = 0.106$) were slightly lower than the Pearson correlation coefficient but consistent in direction.

Table 3: Spearman and Kendall rank correlation coefficients

Correlation Type	Mussel vs *Polysiphonia*	Mussel vs Barnacle	*Polysiphonia* vs Barnacle
Spearman ρ	-0.4433	-0.0165	0.1390
Kendall τ	-0.3300	-0.0096	0.1061

3.2.3 Correlation Significance Tests

Table 4 summarizes the significance test results for pairwise species correlations. All three correlation coefficients between mussels and *Polysiphonia* reached highly significant levels, with $p < 10^{-4}$. The Spearman correlation had the smallest p-value at 4.86×10^{-9}, fully demonstrating a robust negative correlation between the two.

The Pearson correlation between mussels and barnacles was significant at the 0.05 level, $p = 0.046$, but Spearman and Kendall correlations were not significant, with $p > 0.80$, indicating a weak linear relationship between them without monotonicity.

The Pearson correlation between *Polysiphonia* and barnacles was significant at the 0.05 level, $p = 0.029$, but the p-values for Spearman and Kendall correlations approached 0.10 and did not reach the traditional significance level. This suggests there may be a weak positive correlation between them, but the relationship is not robust.

Table 4: Pairwise species correlation significance tests

Species Pair	r	p_{Pearson}	ρ	p_{Spearman}	τ	p_{Kendall}
Mussel vs *Polysiphonia*	-0.3000	1.22×10^{-4}	-0.4433	4.86×10^{-9}	-0.3300	4.98×10^{-8}
Mussel vs Barnacle	-0.1584	0.0461	-0.0165	0.8364	-0.0096	0.8730
Polysiphonia vs Barnacle	0.1730	0.0293	0.1390	0.0805	0.1061	0.0908

3.3 Partial Correlation Analysis Results

Table 5 shows partial correlation coefficients and their significance after controlling for the influence of the third species. After controlling for barnacle influence, the partial correlation coefficient between mussels and *Polysiphonia* was -0.28, $p = 0.00036$, remaining highly significant. Compared to the simple Pearson correlation $r = -0.30$, the absolute value of the partial correlation coefficient decreased slightly, but the difference was small, indicating that the confounding effect of barnacles on the mussel-*Polysiphonia* relationship is minor, and the negative correlation between them is mainly a direct effect.

After controlling for *Polysiphonia* influence, the partial correlation coefficient between mussels and barnacles was -0.11, $p = 0.156$, not significant. After controlling for mussel influence, the partial correlation coefficient between *Polysiphonia* and barnacles was 0.13, $p = 0.095$, also not significant. These results indicate that after excluding the influence of other species, direct correlations between mussels and barnacles, and between *Polysiphonia* and barnacles are weak.

Table 5: Partial correlation coefficients and p-value matrix

	Mussel	*Polysiphonia*	Barnacle
Partial correlation coefficients			
Mussel	1.0000	−0.2803	−0.1134
Polysiphonia	−0.2803	1.0000	0.1332
Barnacle	−0.1134	0.1332	1.0000
p-values			
Mussel	–	0.00036	0.15596
Polysiphonia	0.00036	–	0.09535
Barnacle	0.15596	0.09535	–

3.4 Visualization of Data Distribution Characteristics

Figure 1 shows histograms, boxplots, and kernel density curves for the coverage of the three benthic organisms. From the histograms, it can be seen that the coverage distributions of all three species are right-skewed, with a large number of sampling sites having zero or near-zero coverage, and a few sites having relatively high coverage. Boxplots show that mussels have higher median and interquartile range than *Polysiphonia* and barnacles, with many high-value outliers. Kernel density curves further reveal the peak and long-tail characteristics of the distributions.

The distributions of *Polysiphonia* and barnacles are more extreme, with numerous zero values causing density curves to peak at zero and then decline rapidly. This distribution pattern conforms to the typical characteristics of sessile organisms: under suitable environmental conditions, species can aggregate in large numbers, but at unsuitable sites, they are completely absent.

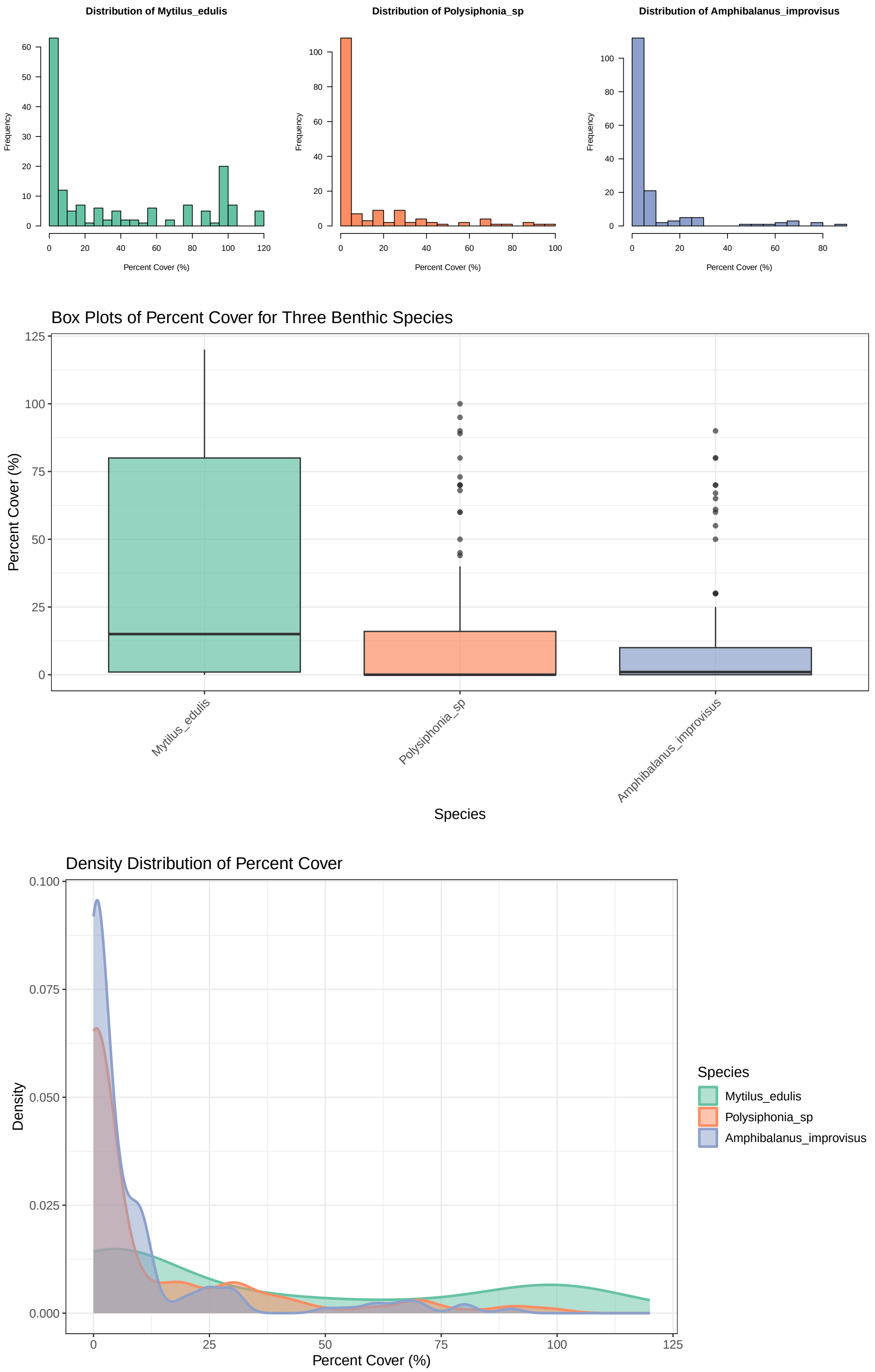

Figure 1: Distribution characteristics of coverage for three benthic organisms: (a) Histograms; (b) Boxplots; (c) Kernel density curves

3.5 Visualization of Interspecific Relationships

Figure 2 shows scatterplot matrices and scatterplots with regression lines for pairwise species. In the scatterplot between mussels and *Polysiphonia*, a clear negative trend can be observed: when mussel coverage is high, *Polysiphonia* coverage tends to be low, and vice versa. The negative slope of the regression line and the confidence band clearly reflect this negative correlation.

Scatterplots between mussels and barnacles, and between *Polysiphonia* and barnacles show relatively dispersed point clouds with no obvious linear trend. A large number of data points are concentrated in the low coverage region, indicating sparse occurrence of barnacles. The `ggpairs` plot shows a comprehensive view of scatterplots, correlation coefficients, and density distributions, further confirming the above relationships.

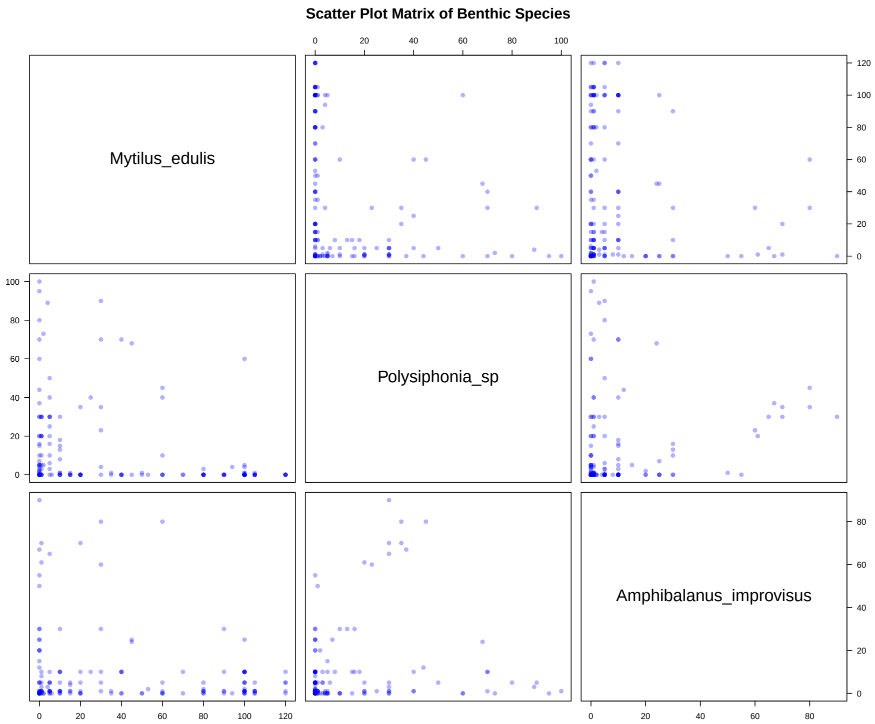

Figure 2: Scatterplot matrix of pairwise species coverage

3.6 Correlation Heatmaps

Figure 5 shows heatmap and `corrplot` visualization of Pearson correlation coefficients. The heatmap uses color gradients to represent the magnitude and direction of correlation coefficients: blue represents positive correlation, red represents negative correlation, and color depth reflects correlation strength. The red block between mussels and *Polysiphonia* is most prominent, intuitively reflecting the significant negative correlation between them. Dark blue on the diagonal represents perfect positive correlation of species with themselves ($r = 1$).

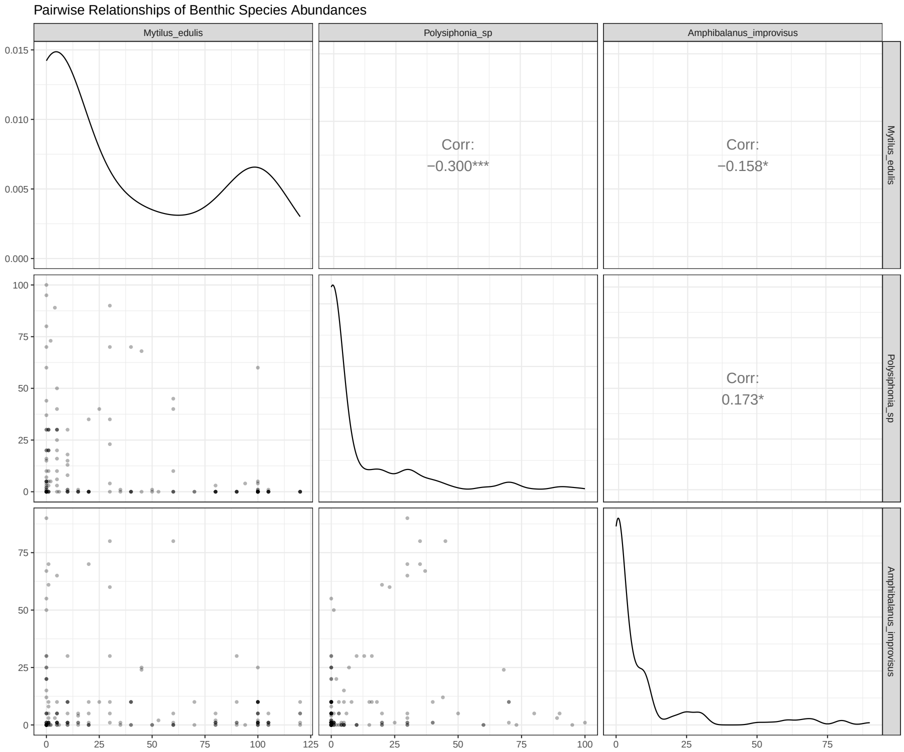

Figure 3: GGally scatterplot matrix of species coverage, including correlation coefficients and density distributions

`corrplot` uses dual encoding of circle size and color for correlation: the larger the circle, the stronger the correlation; color indicates direction. This visualization method is concise and clear, facilitating quick identification of patterns of interspecific relationships.

3.7 Linear Regression Model Results

Table 6 summarizes the fitting results of three groups of simple linear regression models. In the "*Polysiphonia* ~ Mussel" model, the slope was -0.159, $p = 0.00012$, significantly different from zero, indicating that for every 1% increase in mussel coverage, *Polysiphonia* coverage decreased by an average of 0.159%. The coefficient of determination $R^2 = 0.090$, indicating that mussel coverage can explain 9% of the variation in *Polysiphonia* coverage. Although not high, this has practical significance in ecological research.

The slope of the "Barnacle ~ Mussel" model was -0.068, $p = 0.046$, significant at the 0.05 level, but R^2 was only 0.025, with very low explanatory power. The slope of the "Barnacle ~ *Polysiphonia*" model was 0.138, $p = 0.029$, with $R^2 = 0.030$, also having weak explanatory power.

Residual diagnostics showed that QQ plots for all models basically followed the diagonal line, indicating that residuals were approximately normally distributed. Residuals vs. fitted values plots showed some degree of heteroscedasticity, especially in the low fitted value region, but for percent cover data, this phenomenon is quite common.

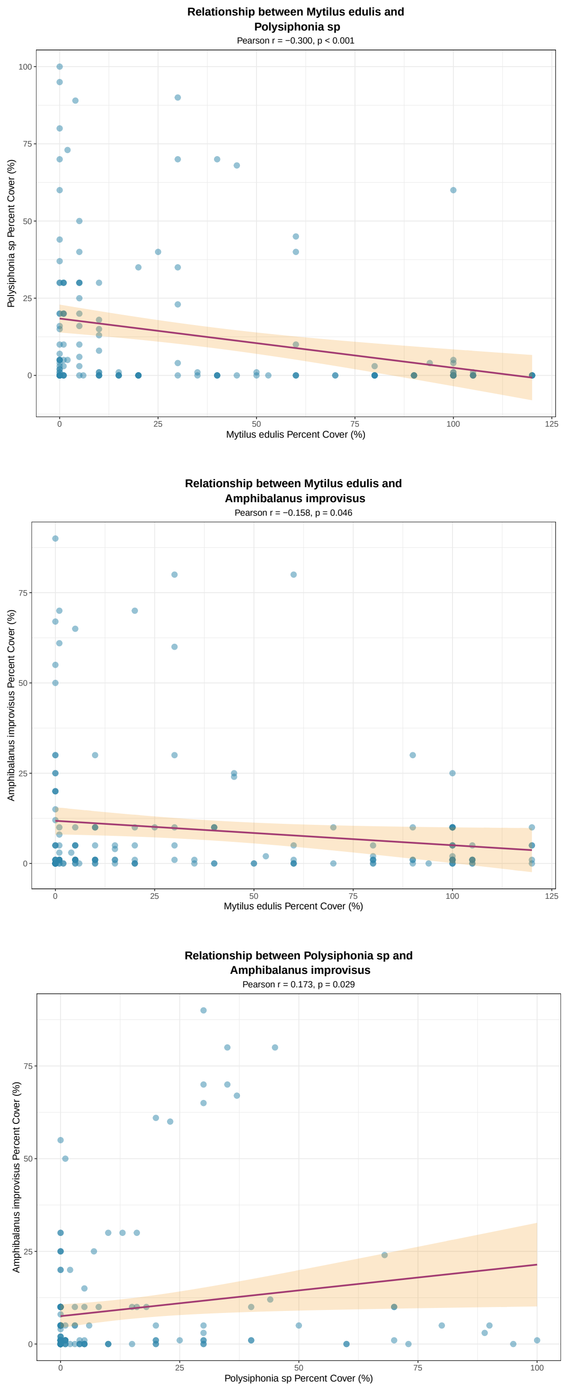

Figure 4: Scatterplots and linear regression fits for pairwise species relationships: (a) Mussel vs *Polysiphonia*; (b) Mussel vs Barnacle; (c) *Polysiphonia* vs Barnacle. Shaded areas represent 95% confidence intervals

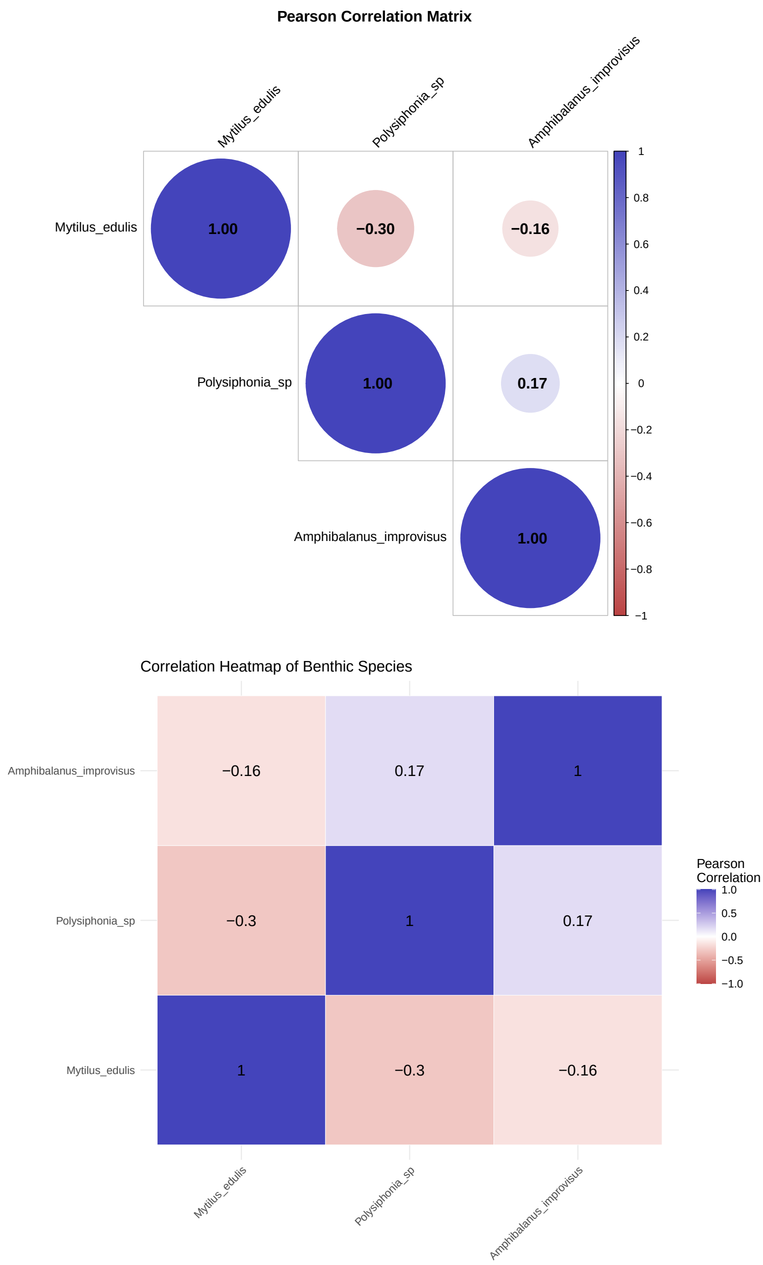

Figure 5: Visualization of Pearson correlation coefficients: (a) `corrplot` circle plot; (b) Heatmap

Log transformation could improve heteroscedasticity issues, but to maintain ecological interpretability, this study retained analysis on the original scale.

Table 6: Linear regression model fitting results

Model	Intercept	Slope	R^2	p-value
Polysiphonia ~ Mussel	18.39	−0.159	0.090	0.00012
Barnacle ~ Mussel	15.28	−0.068	0.025	0.046
Barnacle ~ *Polysiphonia*	7.53	0.138	0.030	0.029

4 Discussion

4.1 Ecological Interpretation of Interspecific Competition Relationships

4.1.1 Negative Correlation Between Mussels and *Polysiphonia*

This study found a robust negative correlation between mussels and *Polysiphonia*, a result confirmed across all three correlation analysis methods and verified as a direct effect through partial correlation analysis. This negative correlation may stem from the following ecological mechanisms:

First, **spatial competition** is the most direct explanation. Both mussels and *Polysiphonia* require attachment to hard substrates for growth. When mussels attach in large numbers and form dense mussel beds, they physically occupy substantial space, reducing the substrate area available for *Polysiphonia* attachment. Conversely, when *Polysiphonia* grows abundantly and forms thick algal mats, its filamentous structure may hinder mussel larval attachment and colonization. This direct spatial exclusion is a common form of competition in sessile organism communities.

Second, **resource competition** may also play an important role. Although mussels are heterotrophic organisms (filter feeders) and *Polysiphonia* is autotrophic (photosynthetic), they may compete for inorganic nutrients (such as nitrogen and phosphorus). While mussel excreta can provide nutrients for algae, the presence of large numbers of mussels also consumes dissolved oxygen in the water, and the organic debris produced may alter local nutrient cycling. These factors may indirectly affect *Polysiphonia* growth.

Third, **microenvironment alteration** may be an important indirect mechanism. Through filter feeding, mussels can significantly change local water plankton concentrations, turbidity, and nutrient levels. Mussel attachment also changes flow patterns and sediment dynamics on substrate surfaces. These microenvironmental changes may negatively impact *Polysiphonia* growth. Similarly, dense algal coverage changes light conditions and oxygen concentrations, which may also affect mussel larval survival.

It is noteworthy that the Spearman rank correlation ($\rho = -0.44$) was stronger than the Pearson linear correlation ($r = -0.30$), suggesting that the competitive relationship between them may not be a simple linear relationship, but rather that effects are more pronounced at certain coverage thresholds. This nonlinear effect may be related to density-dependent competition intensity: when the density of one species exceeds a certain threshold, its inhibitory effect on another species significantly intensifies.

4.1.2 Weak Correlation Between Mussels and Barnacles

Although the Pearson correlation between mussels and barnacles reached significance level, $r = -0.16$, $p = 0.046$, the absolute value of the correlation coefficient was small, and Spearman and Kendall correlations were not significant. Partial correlation analysis also showed no significant direct relationship between them. This indicates that competition between them is relatively weak.

Possible reasons include: First, barnacles may differ from mussels in vertical distribution, flow condition requirements, and reproductive timing, and niche differentiation reduces direct competition. Second, barnacle abundance is mainly determined by larval supply and colonization success rates, which are strongly influenced by environmental factors such as hydrodynamics, temperature, and salinity, possibly masking competitive effects with mussels. Additionally, the study data showed barnacles had a mean coverage of only 9.23%, with many sites having coverage near zero. This low abundance and high variability may limit statistical power to detect interspecific relationships.

4.1.3 Weak Positive Correlation Between *Polysiphonia* and Barnacles

There was a weak positive correlation between *Polysiphonia* and barnacles ($r = 0.17$, $p = 0.029$), but partial correlation was not significant. This weak positive correlation may reflect their preference for similar environmental conditions rather than a direct mutualistic relationship. For example, both may prefer environments with strong currents and abundant nutrients, or they may both show higher abundances at sites with low mussel coverage.

Another possible explanation is **facilitation**: algal photosynthesis produces oxygen, which may improve local oxygen conditions; algae can also provide microhabitats for barnacle larvae attachment. However, the strength of this positive correlation is weak and not robust, requiring further experimental research for validation.

4.2 Ecological Significance of Data Distribution Characteristics

Coverage distributions of all three species were highly right-skewed, with coefficients of variation exceeding 100%, numerous sites having zero or very low coverage, and a few sites having very high coverage. This distribution pattern is typical of the patchy distribution characteristics of benthic sessile organisms, reflecting multiple ecological processes.

First, habitat heterogeneity leads to large differences in environmental conditions such as substrate type, water depth, current, light, and salinity among different sites. Some sites are very suitable for certain species growth, while other sites are completely unsuitable. Second, mussel and barnacle larvae float in water for a period before attaching and settling. This process is influenced by multiple random factors such as currents, distance from larval supply sources, and timing of settlement, resulting in high spatial variability in adult distributions. Third, positive feedback mechanisms promote further attachment of conspecific individuals once a species successfully colonizes a site, forming aggregated distributions.

This distribution pattern has important implications for ecological monitoring and management: sampling point numbers and spatial coverage need to be increased to comprehensively capture benthic organism distribution characteristics; mean-based analyses may mask important ecological processes, requiring comprehensive assessment combining median and variability indicators.

4.2.1 Comparison of Correlation Analysis Methods

This study employed three correlation analysis methods simultaneously: Pearson, Spearman, and Kendall, each with advantages and disadvantages:

Pearson correlation assumes variables have a linear relationship, is sensitive to outliers, and is suitable for normally distributed continuous variables. In this study, percent cover data were right-skewed and did not satisfy normality assumptions, but Pearson correlation remains an effective indicator for measuring linear relationship strength.

Spearman and Kendall correlations are nonparametric methods with no requirements for data distribution, insensitive to outliers, and capable of capturing monotonic but nonlinear relationships. In this study, Spearman correlation coefficients were generally higher than Pearson correlation, indicating some nonlinear components in interspecific relationships. Kendall correlation is more robust for data with smaller sample sizes but is computationally intensive.

Comprehensive use of multiple methods can more fully reveal interspecific relationships and improve the robustness of conclusions. When multiple methods yield consistent conclusions (such as the negative correlation between mussels and *Polysiphonia*), we have more confidence in the existence of this relationship.

4.2.2 Value of Partial Correlation Analysis

Partial correlation analysis reveals direct relationships between variable pairs by controlling for the influence of a third variable, which is particularly important in multispecies system research. Simple correlations may be influenced by confounding factors: for example, two species may show correlation because they jointly respond to a certain environmental factor, but in reality there is no direct interaction between them.

In this study, partial correlation analysis showed that the negative correlation between mussels and *Polysiphonia* remained significant after controlling for barnacle influence, confirming a direct competitive effect between them. The partial correlations between mussels and barnacles, and between *Polysiphonia* and barnacles were not significant, indicating that their simple correlations may be mainly driven by indirect effects or common environmental factors.

4.2.3 Limitations of Linear Regression

The explanatory power of simple linear regression models was low ($R^2 < 0.10$), which is not uncommon in ecological research. Biological abundance is influenced by multiple factors, including environmental conditions, larval supply, predation pressure, and disease. The abundance of a single species can only explain a small portion of the variation in the target species.

Additionally, the zero-inflated characteristics and heteroscedasticity of percent cover data violate linear regression assumptions. More suitable methods may include generalized linear models (such as negative binomial regression), zero-inflated models, or quantile regression. Log transformation can improve heteroscedasticity but changes the interpretation of correlations (from additive to multiplicative) and requires special treatment of zero values.

Nevertheless, linear regression still provides quantitative descriptions of relationship direction and approximate strength, and residual diagnostics help us understand model applicability and limitations.

4.2.4 Data Limitations

This study was based on cross-sectional data and lacked temporal dimension information. Correlation analysis can only reveal associations among species and cannot directly prove causal relationships. To confirm competitive relationships, long-term monitoring data or controlled experiments are needed. For example, through removal experiments that artificially remove a species and observe responses of other species, competitive effects can be more directly assessed.

The data lacked environmental factor information (such as temperature, salinity, currents, nutrients, etc.). These factors may be important drivers of species abundance and may also regulate the intensity of interspecific interactions. Future research should incorporate environmental covariates and use multivariate statistical methods (such as multiple regression, partial least squares, structural equation modeling) to separate the effects of environmental factors and biological interactions.

Although the sample size reached 159, considering the high heterogeneity and zero-inflation characteristics of the three species' distributions, increasing sample size would still help improve statistical power. Additionally, information on spatial distribution and temporal span of samples was not reflected in the data, limiting analysis of spatial patterns and temporal dynamics.

4.2.5 Improvement of Statistical Methods

In addition to the aforementioned generalized linear models and zero-inflated models, future research could consider the following methods:

(1) **Spatial statistical methods:** If spatial coordinate information for samples is available, methods such as spatial autocorrelation analysis (Moran's I, Geary's C), variogram analysis, and spatial regression models can be used to consider the influence of spatial dependence on species relationships.

(2) **Hierarchical models:** If monitoring spans multiple years or locations, mixed-effects models can be used, treating year or location as random effects to assess the consistency of interspecific relationships across different spatiotemporal scales.

(3) **Network analysis:** Multiple species (including *Ceramium* and others not covered in this study) can be included in analysis to construct species co-occurrence networks and identify key species and modular structure in communities.

(4) **Machine learning methods:** Methods such as random forests and gradient boosting trees can capture complex nonlinear relationships and interactions. Although less interpretable, they may have stronger predictive power.

4.2.6 Further Research on Ecological Mechanisms

Correlation analysis provides statistical evidence of interspecific relationships but cannot directly reveal ecological mechanisms. To deeply understand the competitive mechanisms between mussels and *Polysiphonia*, the following research is needed:

(1) **Field manipulation experiments:** Set up different density gradient treatments of mussels and *Polysiphonia* in natural environments and monitor their growth, survival, and reproductive responses to each other.

(2) **Laboratory controlled experiments:** Study specific mechanisms such as spatial competition, nutrient competition, and allelopathy under controlled conditions.

(3) **Functional trait analysis:** Measure functional traits such as growth rate, attachment strength, and metabolic rate of species, and understand competition intensity in combination with trait differences.

(4) **Long-term monitoring:** Establish fixed quadrats for multi-year continuous monitoring, record temporal dynamics of species abundance and environmental conditions,

and use time series analysis methods (such as Granger causality test) to assess dynamic relationships among species.

4.3 Implications for Coastal Zone Management

The results of this study have practical significance for monitoring and management of benthic communities along the Baltic Sea coast:

(1) **Focus on interactions among key species:** The negative correlation between mussels and *Polysiphonia* indicates that management measures (such as mussel harvesting and invasive species control) may indirectly affect other species by changing the abundance of one species. Interspecific relationships should be considered when formulating management strategies.

(2) **Identify areas with high competitive pressure:** In areas with particularly high mussel density, *Polysiphonia* growth may be strongly inhibited. If *Polysiphonia* has important ecological functions (such as serving as food for certain herbivores or providing habitat), measures may need to be taken to maintain its population.

(3) **Selection of monitoring indicators:** The high variability of the three species suggests that single sampling may not represent the true state of the community. It is recommended to increase sampling frequency and spatial coverage and use indicators such as coefficient of variation and quantiles to supplement mean information.

(4) **Predict impacts of environmental changes:** Climate change may alter species growth rates, reproductive cycles, and physiological adaptability, thereby changing the competitive balance among species. For example, if temperature increases are more favorable for mussel growth, they may further inhibit *Polysiphonia*, leading to community structure homogenization. Accumulation of monitoring data helps predict and adapt to these changes.

(5) **Considerations for ecological restoration:** When conducting benthic habitat restoration, interspecific interactions should be considered. Simply increasing one species (such as releasing mussel seedlings) may have negative impacts on other species and requires comprehensive assessment.

5 Conclusions

Through multiple statistical methods, this study systematically analyzed the abundance correlations among three dominant benthic organisms along the Schleswig-Holstein coast of the Baltic Sea, reaching the following main conclusions:

1. Coverage distributions of mussels, *Polysiphonia*, and barnacles were all highly dispersed and right-skewed, reflecting typical patchy distribution characteristics of benthic sessile organisms. Coefficients of variation for all three species exceeded 100%,

indicating important influences of habitat heterogeneity and random colonization processes on community structure.

2. There was a robust negative correlation between mussels and *Polysiphonia*, a conclusion consistently supported by Pearson ($r = -0.30$), Spearman ($\rho = -0.44$), and Kendall ($\tau = -0.33$) correlation analyses, all reaching highly significant levels ($p < 10^{-4}$). Partial correlation analysis showed this negative correlation was mainly a direct effect, confirming a competitive relationship between them.

3. Spearman rank correlation was stronger than Pearson linear correlation, suggesting that the competitive relationship between mussels and *Polysiphonia* may have nonlinear components, with competition intensity possibly having a nonlinear relationship with species density.

4. Correlations between mussels and barnacles, and between *Polysiphonia* and barnacles were weak and not robust. Although simple correlations reached significance under the Pearson method, rank correlations were not significant, and partial correlation analysis found no direct relationships, indicating weak interactions between these species pairs.

5. Linear regression analysis showed that mussels could explain 9% of the variation in *Polysiphonia* coverage. Although explanatory power was limited, this has practical significance in highly complex ecosystems. Residual diagnostics indicated some heteroscedasticity in models, which could be improved in the future using generalized linear models or zero-inflated models.

6. This study provides a complete, reproducible data analysis workflow, including multiple statistical methods and visualization techniques, offering methodological reference for similar ecological studies. All analysis code and results are publicly available, ensuring research transparency and reproducibility.

This study reveals patterns of interactions among dominant species in Baltic Sea benthic communities, providing quantitative evidence for understanding community structure and dynamics. The competitive relationship between mussels and *Polysiphonia* suggests that comprehensive consideration of interspecific interactions is needed in benthic organism management and conservation. Future research should incorporate environmental factors, long-term monitoring data, and experimental validation to further elucidate the ecological mechanisms of competition and their impacts on community stability and ecosystem functioning.

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